New Imperfect High Threshold and Slice Secret Sharing Schemes

Andrej Bogdanov, Yanbo Chen

University of Ottawa

Threshold secret sharing (for 1 bit secret)

- Share: $s \in \{0,1\} \rightarrow x_1, \ldots, x_n$
- Reconstruction: Any t parties can recover s
- (Perfect) privacy: The view of any t-1 parties has the same distribution for secrets 0 and 1
- Weak privacy [Beimel-Franklin'07]: Any t-1 parties can't rule out the possibility of either secret

2-out-of-n weakly private scheme [BF'07]

- Shares of secret 0:
  - ...,1,1,2,1,1,... (A random party receives 2, others receive 1)
  - ...,3,3,4,3,3,...
- Shares of secret 1:
  - ...,2,2,3,2,2,...
  - ...,4,4,1,4,4,...
- Reconstruction: two parties' view
  - Secret 0: (1,1), (1,2), (3,3), (3,4)
  - Secret 1: (2,2), (2,3), (4,4), (1,4)
- Weak privacy: one party's view can be 1,2,3,4 for both secrets
- Alphabet size 4!
- Perfect schemes require alphabet size $\Omega(n)$ [Kilian-Nisan'90,Bogdanov-Guo-Komargodski'16]

Duality

- A phenomenon with surprising predictive power: Best share size for $A \approx$ best share size for its dual A^*
  - t-out-of-n $\Longrightarrow$ (n-t+1)-out-of-n
- [BF'07, Beimel-Ishai-Kushilevitz-Li'25]: Weakly private 2-out-of-n, alphabet size $O(1)$; t-out-of-n, alphabet size $O_t(1)$
  - $\Longrightarrow$ (n-1)-out-of-n?
  - $\Longrightarrow$ (n-d+1)-out-of-n?
- Our main results:
  - (n-1)-out-of-n: Yes!
  - (n-d+1)-out-of-n: Yes under a conjecture
  - And even better…

(n-1)-out-of-n weakly private scheme

- Share alphabet $\mathbb{F}_5$
- Uniformly sample the shares conditioned on $\sum_{i=1}^{n} x_i = 0$, and $H\left(\sum_{i<j} x_i\, x_j\right) = s$
- Reconstruction: $0 = \sum_{i=1}^{n} x_i = x_1 + \sum_{i=2}^{n} x_i$

Privacy

- Missing x_1, x_2
- $0 = \sum_{i=1}^{n} x_i = x_1 + x_2 + \sum_{i=3}^{n} x_i$
- $? = \sum_{i<j} x_i\, x_j = x_1 x_2 + f(x_3 \cdots x_n)(x_1 + x_2) + g(x_3 \cdots x_n)$
- Knowing $x_1 + x_2$, possible values of $x_1 x_2$?
- $(X - x_1)(X - x_2) = X^2 - (x_1 + x_2)X + x_1 x_2$
- $x_1 x_2 = b_0$ can be any value such that $X^2 + b_1 X + b_0$ has a root in $\mathbb{F}_5$
- $X^2 + b_1 X + b_0 = (X + b_1/2)^2 - b_1^2/4 + b_0$ has a root
- $\Leftrightarrow b_1^2/4 - b_0$ is a QR, {0,1,4} for $\mathbb{F}_5$
- The set of possible values of $\sum_{i<j} x_i\, x_j = b_0 + \text{known}$ is a shift of {0,1,4}
- Let $H(0,1,3) = 0, H(2,4) = 1$
- H is nonconstant on any shift of {0,1,4}

(n-d+1)-out-of-n scheme

- Share alphabet $\mathbb{F}_q$
- d-1 constraints up to degree d-1; $H\left(f\left(x_{[n]}\right)\right) = s$, f is of degree d
- Reconstruction: d-1 constraints recover d-1 missing shares
- Privacy: $f\left(x_{[n]}\right) = b_0 + \text{known}$
- b_0 can be any value such that $X^d + b_{d-1}X^{d-1} + \cdots + b_0$ splits completely over $\mathbb{F}_q$
- Conjecture: if q is sufficiently large and d divides $q-1$, then for any $b_{d-1}\ldots, b_1$ there are $\Omega_d(q)$ (roughly $q/d!$) many such b_0 (True for $d = 2,3$)
- For $q = O_d(1/\epsilon^2 \log 1/\epsilon)$, a random H suffices for **ϵ-statistical privacy**
- The conjecture essentially: the condition "d divides $q-1$" rules out the exceptional case of Chebotarev's density theorem

Top slice access structure

- t-out-of-n slice access structure: minimal qualified sets and maximal unqualified sets all have t parties
- Statistical (n-2)-out-of-n slice with $O\big((\log 1/\epsilon)(\log n)\big)$ share size
  - Match previous computationally private scheme [Beimel-Farràs-Lasri-Nir'24]

Summary, and Duality Again

Duality holds?

- Perfect threshold scheme [Bogdanov'23]
- Weakly private threshold scheme:
  - 2-out-of-n, alphabet size 4 [BF'07]
  - (n-1)-out-of-n, alphabet size **4**
  - t-out-of-n, alphabet size $O_t(1)$ [BF'07,BIKL'25]
  - (n-d+1)-out-of-n, alphabet size $O_d(1)$

Duality does not hold

- ϵ-statistical threshold scheme:
  - 2-out-of-n requires alphabet size $\Omega_\epsilon(n)$ [KN'90, Abram-Beimel-Ishai-Kushilevitz-Narayanan'23]
  - (n-1)-out-of-n, alphabet size $O_\epsilon(1)$

?

- ϵ-statistical slice scheme:
  - (n-2)-out-of-n, share size $O_\epsilon(\log n)$
  - Perfect 2-out-of-n, $\exp \tilde{O}\left(\sqrt{\log n}\right)$ [Liu-Vaikuntanathan-Wee'17], improvement with statistical privacy?

Thanks !